

Referring-Aware Visuomotor Policy Learning for Closed-Loop Manipulation

Anonymous CVPR submission

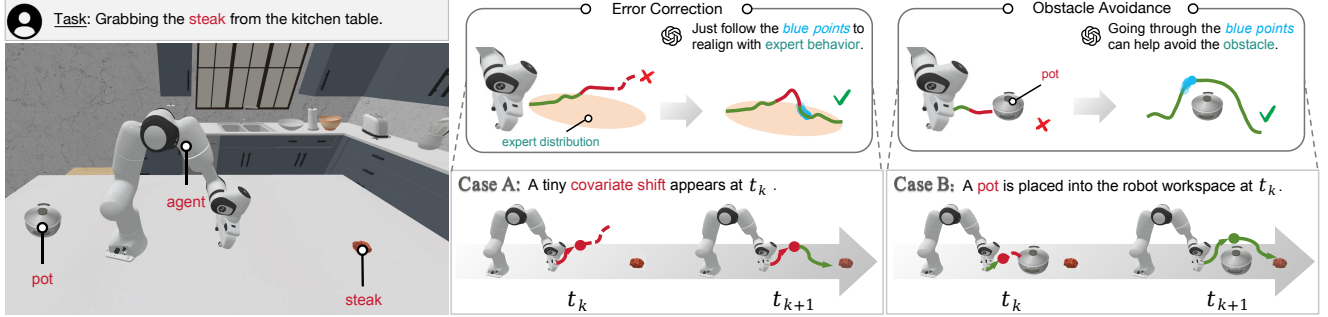


Figure 1. The role of the proposed Referring-Aware Visuomotor Policy (ReV) in mitigating out-of-distribution failures. In executing the task “Grabbing the steak from the kitchen table”, several challenges can be encountered. **Case A:** a small covariate shift at t_k quickly leads to a compounding misalignment with expert demonstrations, driving the embodied agent into unseen states and resulting in task failure. **Case B:** an unforeseen obstacle (e.g., a pot) emerges at t_k to dynamically block the robot’s path, leading to failure as well. Unlike traditional imitation learning methods that often struggle to generalize to unseen states or observations, the proposed ReV employs coupled diffusion heads to online react to external sparse referring provided by a human or a high-level reasoning planner. Without requiring additional training data or complex fine-tuning in post-processing, the model can effectively address Case A and Case B in real-world applications.

Abstract

This paper addresses a fundamental problem of visuomotor policy learning for robotic manipulation: how to enhance robustness in out-of-distribution situations (i.e., execution errors or dynamically re-routing trajectories) where the model relies solely on the original expert demonstrations for training. We introduce the **Referring-Aware Visuomotor Policy (ReV)**, a closed-loop framework capable of responding to unforeseen circumstances through the instant incorporation of sparse referring points provided by a human or a high-level reasoning planner. Specifically, ReV is a newly designed policy model with coupled diffusion heads to preserve standard task execution patterns while seamlessly integrating sparse referring via a trajectory-steering strategy. Upon receiving a specific referring point, the global diffusion head firstly generates a sequence of globally consistent yet temporally sparse action anchors, while identifies the precise temporal position for the referring point within this sequence. Subsequently, the local diffusion head is updated to generate a dense action trajectory between adjacent anchors. This closed-loop process repeats at every execution step, enabling real-time trajectory replanning in response to dynamic changes in the scene. In practice, rather than

relying on elaborate annotations, ReV is trained only by applying targeted perturbations to expert demonstrations. Without any additional data or fine-tuning scheme, ReV improves success rates on a suite of challenging simulated benchmarks and consistently outperforms prior “condition-and-generate” baselines in real-world environments.

1. Introduction

With the development of large-scale simulated and real-world robotic datasets [4, 20, 21, 34], imitation learning based visuomotor policy models [2, 28, 40, 41] have shown the ability to perform a wide range of daily tasks. However, the imitation objective of these visuomotor policy models provides no explicit mechanism for recovery. Consequently, while these data-driven visuomotor policies excel at executing instructed tasks, they become very fragile to out-of-distribution situations once states and observations drift outside the demonstrated distribution. As shown in Fig. 1, they cannot recover from execution errors or replan trajectories to choose safer, more reasonable paths.

To address such an issue, one possible solution is to expand the training distribution with large datasets of er-

044	rors and human corrections, explicitly training the policy	097
045	model to recover from errors [8, 16, 17]. However, these	098
046	approaches require enormous human effort, do not scale	099
047	well, and may even compromise success rates by intro-	100
048	ducing suboptimal trajectories. Another line of work re-	
049	lies on open-loop reasoning planners [12] to address <i>out-</i>	
050	<i>of-distribution</i> scenarios, drawing on the strong priors of	
051	high-level models, such as large language models. These	
052	methods first generate intermediate robot poses, which are	
053	then executed by rule-based motion or grasping modules.	
054	Although they introduce reasoning into the manipulation,	
055	their low-frequency rule-based interactions still make them	
056	ill-suited in dynamic environments. Given this, one ques-	
057	tion naturally arise: <i>when limited to expert demonstration</i>	
058	<i>data for imitation learning, how can we enhance the robust-</i>	
059	<i>ness of the policy model in out-of-distribution situations?</i>	
060	In this paper, we introduce a referring-awareness visuo-	
061	motor policy model termed ReV, which is a closed-loop	
062	framework that can incorporate external referring informa-	
063	tion (e.g., from humans or high-level reasoning planners) to	
064	enhance both adaptability and generalization. By receiving	
065	referring information into carefully designed architecture,	
066	ReV enables the model to flexibly handle error recovery	
067	(leading the robot back to the expert distribution) or per-	
068	form precise goal-oriented adjustments (navigating to safer	
069	or more optimal regions), without requiring additional train-	
070	ing data or complex fine-tuning in post-processing.	
071	In practice, ReV leverages 3D referring points for guid-	
072	ance. Compared with the language instructions, this rep-	
073	resentation provides sub-centimeter spatial precision while	
074	avoiding the linguistic ambiguities that typically challenge	
075	visuomotor policies. We further introduce a novel archi-	
076	tecture for ReV that employs coupled diffusion heads	
077	to achieve a more effective response to referring points.	
078	Specifically, upon receiving a referring point, a Temporal-	
079	Position Prediction module is firstly adopted to estimate its	
080	plausible location along the execution trajectory. Then the	
081	trajectory-steering strategy feeds this temporally-positioned	
082	point into the Global Diffusion Head, producing a series	
083	of sparse but precise action anchors that reliably reach the	
084	specified targets. Subsequently, a Local Diffusion Head	
085	conducts temporal-dependent interpolation strategies be-	
086	tween consecutive anchors, progressively refining them into	
087	a smooth and fine-grained trajectory. The full model oper-	
088	ates recurrently at each inference step, enabling dynamic	
089	online replanning in response to evolving scene conditions.	
090	The main contributions can be summarized as follows.	
091	1) We present a referring-aware visuomotor policy that op-	
092	erates within an imitation learning framework. By integrat-	
093	ing point-level referring cues with task-specific execution	
094	patterns, it enables embodied agents to effectively handle	
095	challenging <i>out-of-distribution</i> scenarios. 2) We introduce	
096	a novel policy model with coupled diffusion heads that gen-	
	erates actions in a coarse-to-fine manner, well supporting	097
	closed-loop inference. 3) Extensive experiments in both	098
	simulation and real-world settings show that ReV achieves	099
	SOTA performance in referring-aware manipulation tasks.	100
	2. Related Works	101
	2.1. Visuomotor Policy Models for Manipulation	102
	Visuomotor policy models integrate visual perception with	103
	motor control, enabling robots to perform manipulation	104
	tasks in complex and unstructured environments [1, 10, 22,	105
	27, 35, 43]. Two generative paradigms have recently gained	106
	significant attention for addressing challenges in this do-	107
	main. Autoregressive models [6, 9, 14, 26, 39, 45, 46] de-	108
	compose the trajectory distribution into a sequence of next-	109
	step conditionals. This factorization enables efficient train-	110
	ing and fast inference at deployment. However, this token-	111
	by-token generation process lacks the ability to revise ear-	112
	lier decisions, causing small deviations to accumulate and	113
	degrade global coherence over time. Recently, diffusion	114
	models [11, 29], have proven remarkably effective for tra-	115
	jectory synthesis [5, 15, 30, 36, 37, 44]. Their ability to	116
	model intricate, multimodal distributions lets them generate	117
	more accurate and flexible robot motions. However, these	118
	methods struggle to handle <i>out-of-distribution</i> situations be-	119
	cause of their reliance on imitation learning.	120
	2.2. Inference-Time Policy Steering	121
	Recent studies have increasingly focused on inference-time	122
	steering of pre-trained robotic policies, aiming to boost task	123
	performance without additional data collection or weight	124
	updates. However, existing methods universally operate	125
	at the single-action or short-horizon level: value-based re-	126
	rankers sample multiple one-step candidates and retain the	127
	one with the highest Q-value [18]; human-in-the-loop steer-	128
	ing translates a human-provided sub-goal into an imme-	129
	diate action offset [38]; model-predictive or safety-driven	130
	schemes unroll only a handful of future steps, iteratively	131
	nudging the next command to respect physical constraints	132
	or avoid obstacles [7, 19, 23, 31, 48]. We instead extract the	133
	global execution patterns for each task from expert demon-	134
	stration trajectory, and then conduct policy steering from a	135
	trajectory-level perspective, yielding a trajectory that bal-	136
	ances the steering cue with task fidelity.	137
	3. Methodology	138
	In the literature, Diffusion Policy [5] imitates expert demon-	139
	strations for low-level execution, yet its “condition-and-	140
	generate” paradigm cannot reliably respond to the exter-	141
	nal reasoning model. Our goal is to develop a diffusion-	142
	based policy that can effectively incorporate external guid-	143
	ance—provided by humans or high-level reasoning plan-	144

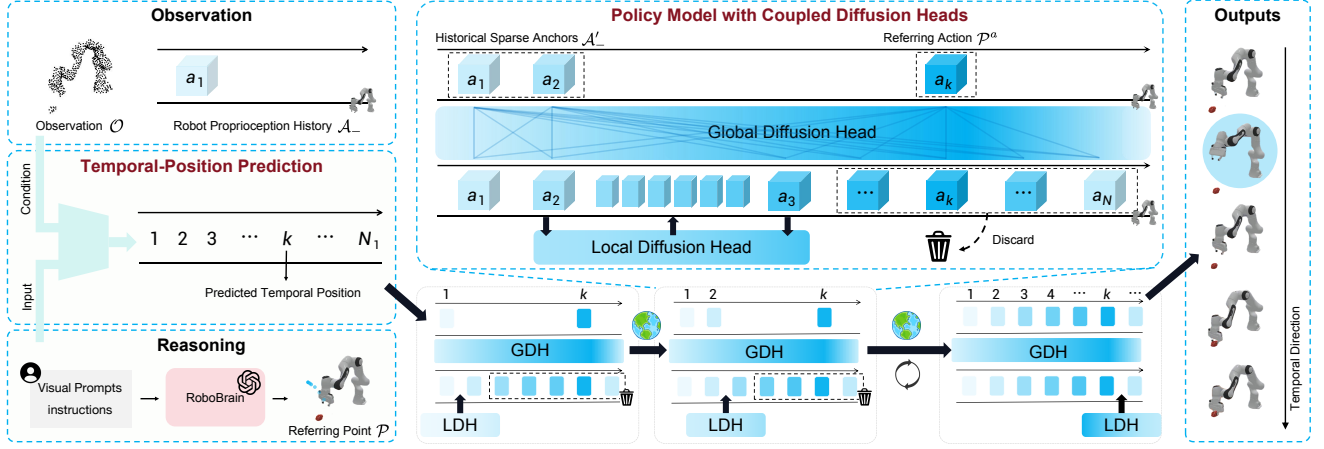


Figure 2. **Overview of our Referring-Aware Visuomotor Policy.** Given the observation $\mathcal{O}$ and robot proprioception history $\mathcal{A}_-$, the *Temporal-Position Prediction Module* on the left assign the optimal temporal position k for the referring point $\mathcal{P}$, and *Policy Model with Coupled Diffusion Heads* on the right injects this temporal-positioned referring point, generating the final trajectory $\mathcal{A}$ satisfying the *Manipulation Steering Choreography*. Specifically, at step i , the global diffusion head (GDH) incorporates both current sparse anchor history $\mathcal{A}'_-$ and $\mathcal{P}$ via the trajectory-steering strategy to yield a sparse yet globally consistent anchor sequence $\mathcal{A}' = \{a'_j\}_{j=1}^{N_1}$, in which $\{a'_j\}_{j=1}^{i-1}$ is anchor history $\mathcal{A}'_-$, $\{a'_j\}_{j=i}^{i+1}$ is interpolated by the local diffusion head (LDH) through a learnable, temporal-dependent interpolation strategy, and $\{a'_j\}_{j=i+1}^{N_1}$ is discarded and will be updated at upcoming inference step. At inference step $i+1$, the anchor history $\mathcal{A}'_-$ is updated by appending the newly generated anchor a'_{i+1} at step i , and the entire policy model repeats, continuously updating $\mathcal{A}$.

ners—and generate trajectories online that accurately satisfy complex spatial constraints specified in the guidance.

3.1. Problem Formulation

We consider the problem of learning a diffusion-based policy π_θ . Given a 3D referring point $\mathcal{P}$ (from a high-level reasoning planner such as RoboRefer [47]), the current visual observation $\mathcal{O}$ and the robot proprioception history $\mathcal{A}_-$, π_θ generates a trajectory $\mathcal{A}$ satisfying the criteria collectively termed *Manipulation Steering Choreography*.

$$\pi_\theta(\mathcal{P}, \mathcal{O}, \mathcal{A}_-) \rightarrow \mathcal{A} \quad (1)$$

Manipulation Steering Choreography. Guided by the high-level reasoning planner, the end-effector must glide through the 3D referring point $\mathcal{P}$ while guaranteeing successful task completion. Thus, we require every generated trajectory to satisfy: 1) *Steering Fidelity*: the trajectory must establish contact within the spatial region nearby the specific 3D referring point. 2) *Task Success*: the trajectory must accomplish the designated manipulation task. 3) *Smoothness*: the end-effector pose must vary uniformly, yielding low-jerk transitions along the entire trajectory.

Referring-Aware Policy Model. To effectively handle the *out-of-distribution* situations, we propose ReV, described below in two parts. **Part 1:** We first build a *policy model with coupled diffusion heads* to learn standard task execution patterns from demonstrations: its global diffusion head (GDH) produces sparse action anchors encoding long-range

motion intent, while the local diffusion head (LDH) interpolates these anchors into a fine-grained trajectory. During inference, the entire model is invoked repeatedly, enabling the robot to replan as the scene changes. **Part 2:** Subsequently, we embed a *referring-aware design* into the aforementioned model, which centers on a temporal-position prediction module as shown in Fig. 2 and a trajectory-steering strategy. The former assigns a temporal position along the entire trajectory to the 3D referring point based on the robot current observation and state history. And the latter injects this temporally-positioned point into our policy model, enabling the generation of a trajectory that fulfills the *Manipulation Steering Choreography*.

3.2. Policy Model with Coupled Diffusion Heads

While previous methods [5, 15, 44, 46] model “observation-policy” coupling within a sliding-window horizon, they fail to capture long-range dependencies across the entire task trajectory. Such a scheme disrupts trajectory consistency and makes it difficult to schedule the referring points $\mathcal{P}$ along the entire manipulation trajectory. To overcome this limitation, we propose a policy model with coupled diffusion heads which can learn standard execution pattern from expert demonstration in a close-loop manner.

Coupled Diffusion Heads. In robotic manipulation, trajectories are prohibitively long, and their length grows with task complexity. Learning a single Diffusion Policy that captures execution pattern of an entire trajectory is therefore

impractical. In this paper, we introduce a coupled diffusion heads: A GDH is first deployed to generate a globally consistent yet temporally sparse action anchors $\mathcal{A}' = \{a'_i\}_{i=1}^{N_1}$ of length N_1 . As illustrated in Fig. 2, GDH predicts $\mathcal{A}'$ conditioned on the current observation $\mathcal{O}$ and the previously executed anchors $\mathcal{A}'_-$, i.e.,

$$\mathcal{A}' \sim f_{\text{GDH}}(\mathcal{A}' | \mathcal{O}, \mathcal{A}'_-) \quad (2)$$

Subsequently, a LDH densifies these anchors, producing a fine-grained trajectory ready for direct deployment on the robot. Following Eq. (2), once the neighboring anchor pair (a'_i, a'_{i+1}) for step i is available, LDH interpolates between them conditioned on the corresponding observation $\mathcal{O}_i$, i.e.,

$$\mathcal{A}_i \sim f_{\text{LDH}}(\mathcal{A}_i | \mathcal{O}_i, a'_i, a'_{i+1}, i) \quad (3)$$

where $\mathcal{A}_i = \{a_{ij}\}_{j=1}^{N_2}$ is the fine-grained sub-trajectory of length N_2 in step i , and the entire manipulation trajectory is simply the concatenation $\mathcal{A} = \{\mathcal{A}_i\}_{i=1}^{N_1}$. It is worth noting that, in Eq. (3), the step index i is explicitly fed into f_{LDH} so that the network can learn temporal-dependent interpolation strategies that vary with the temporal position inside $\mathcal{A}'$. Instead of the conventional “condition-and-generate” paradigm, we apply our trajectory-steering strategy (Sec. 3.3) to (i) historical anchors $\mathcal{A}'_-$ that guides GDH prediction and (ii) the neighboring anchor pair (a'_i, a'_{i+1}) that steers LDH interpolation, ensuring the generated trajectory strictly continues the already-executed motion while fulfilling the remaining task objectives. Since the output of our policy model is a *fixed-length* trajectory, we must select the total horizon $N = N_1 \times N_2$ for each task once at the outset, according to its execution complexity. Fortunately, this mild restriction is easily accommodated in concurrent robotic manipulation pipelines, where an upper bound can be set without impairing deployment.

Closed-Loop Inference. To remain robust to the dynamic environment, we invoke the entire policy model iteratively during inference. As illustrated in Fig. 2, the sparse action anchors $\mathcal{A}'$ is updated online by continuously incorporating the latest observation $\mathcal{O}$ and the ever-growing anchor history $\mathcal{A}'_-$. Specifically, at step i we generate $\mathcal{A}'$ with GDH (Eq. (2)), extract the neighboring anchor pair (a'_i, a'_{i+1}) and produce the fine-grained sub-trajectory $\mathcal{A}_i$ via LDH (Eq. (3)) for immediate execution. Once the corresponding execution process is completed, the executed anchor a'_{i+1} is appended to build the anchor history at step $i+1$, i.e.,

$$\mathcal{A}'_{-,i+1} = \{\mathcal{A}'_{-,i}, a'_{i+1}\} \quad (4)$$

where $\mathcal{A}'_{-,i} = \{a'_j\}_{j=1}^i$ is the anchor history at step i . This $\mathcal{A}'_{-,i+1}$ is then used to update action anchors $\mathcal{A}'$ at step $i+1$.

3.3. Referring-Aware Design

We now detail how to embed the referring-aware design into the aforementioned policy model in order to schedule the

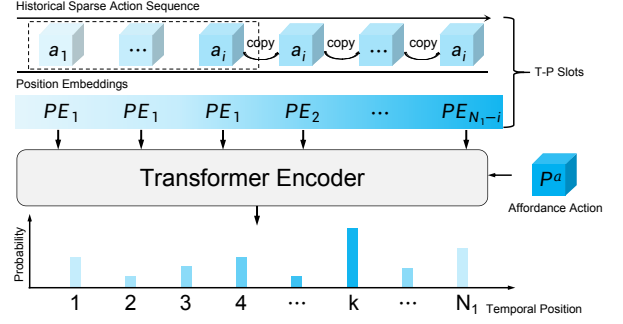


Figure 3. **Temporal-Position Prediction.** The slots buffer $\mathcal{S}$ covers both retained history and extrapolated future steps. Specifically, at step i , we load the historical action anchors $\{a'_j\}_{j=1}^i$ to initialize history part, and the future part is padded by copies of the last action anchor a'_i . Then, the whole slots buffer is augmented with a monotonically-increasing temporal-position embeddings.

referring point $\mathcal{P}$ along the entire trajectory.

Temporal-Position Prediction. As shown in Fig. 3, we formulate the temporal localization of the referring point $\mathcal{P}$ as an N_1 -way classification problem. Concretely, we construct a *fixed-length* timeline by creating a buffer of N_1 temporal-position slots $\mathcal{S}$, covering both retained history and extrapolated future steps. In step i , we build $\mathcal{S}$ as follows: The first $(1, \dots, i)$ slots are loaded with the historical action anchors $\{a'_j\}_{j=1}^i$, and the remaining $(i+1, \dots, N_1)$ slots are padded with copies of the last action anchor a'_i , yielding a slot sequence $[a'_1, a'_2, \dots, a'_i, \dots, a'_i]$. This slot sequence is then augmented with the monotonically-increasing temporal-position embeddings. Specifically, the historical slots $(1, \dots, i)$ share the same temporal-position embedding PE_1 , while the padded slots $(i+1, \dots, N_1)$ receive distinct embeddings $PE_2 < PE_3 < \dots < PE_{N_1-i}$. These embeddings encode the temporal distance from the action in each slot to the referring point $\mathcal{P}$: PE_1 marks the nearest moment, and larger embeddings indicate progressively more distant future times. A transformer encoder processes the augmented $\mathcal{S}$ together with $\mathcal{P}$ and outputs a probability vector $\mathbf{p} = \{p_i\}_{i=1}^{N_1}$ over the entire slots buffer. The predicted temporal position is taken as the slot index with the highest probability, i.e.,

$$k = \arg \max_{0 < i \leq N_1} f_{\text{enc}}(\mathbf{p} | \mathcal{S}, \mathcal{P}) \quad (5)$$

Trajectory-Steering Strategy. Before detailing its mechanics, it is worth noting that all states and actions are represented in *end-effector Cartesian space*, rather than in *joint space*. Formally, the actions considered in this paper are decomposed into: 1) an end-effector pose component $a_{\text{ee}} = (a_{\text{trans}}, a_{\text{rot}})$, where $a_{\text{trans}} \in \mathbb{R}^3$ and $a_{\text{rot}} \in \mathbb{R}^4$ denote the position and rotation respectively. 2) a binary gripper component $a_{\text{gripper}} \in \{0, 1\}$ that opens or closes

the parallel jaw. This representation allows us to convert the constraint on the referring point $\mathcal{P}$ into a *partial* constraint on the referring action $\mathcal{P}^a$: we simply force its translation component a_{trans} to coincide with the referring point $\mathcal{P}$, while leaving the rotation component a_{rot} free to be optimized by the policy model. Borrowing the masked-denoising in [13, 32, 33], we define a binary mask $\mathcal{M}$ ($\mathcal{M}_i = 1$ at temporal position of $\mathcal{P}^a$ predicted via Eq. (5), 0 otherwise) and enforce the denoising process via

$$\mathbf{z}_t = \mathcal{M} \odot \mathcal{P}^a + (1 - \mathcal{M}) \odot \mathbf{z}_t \quad (6)$$

where $\mathbf{z}_t$ is the intermediate results at diffusion denoising timestep t and $\odot$ denotes the Hadamard (element-wise) product. As stated in Sec. 3.2, this strategy is also applied to (i) sparse-anchor generation in GDH and (ii) anchor interpolation in LDH. For GDH we replace $\mathcal{P}^a$ with the anchor history $\mathcal{A}'_-$, and for LDH we replace it with the neighboring pair (a'_i, a'_{i+1}) . To allow our policy model to attend to both the anchor history $\mathcal{A}'_-$ and the referring action $\mathcal{P}^a$, we introduce the referring action $\mathcal{P}^a$ into Eq. (2), i.e.,

$$\mathcal{A}' \sim f_{\text{GDH}}(\mathcal{A}' | \mathcal{O}, \mathcal{A}'_-, \mathcal{P}^a) \quad (7)$$

3.4. Training Strategy

Data Augmentation. To enlarge the effective range of $\mathcal{P}$ and promote generalization, we perform on-the-fly augmentation: we randomly sample an action from the expert demonstrations, perturb it with noise drawn from a broad distribution, and obtain a synthetic referring action $\mathcal{P}^a$. A seventh-order polynomial spline then smoothly blends this synthetic action with its temporal neighborhood, yielding a jerk-bounded trajectory. These augmented trajectories are fed to both GDH and LDH, enabling the system to handle a richer spectrum of referring points $\mathcal{P}$ at inference time.

Training Optimization Process of ReV. The transformer encoder used for temporal-position prediction is trained with the following categorical cross-entropy loss

$$\mathcal{L}_{\text{CCE}} = - \sum_{i=1}^{N_1} y_i \log p_i \quad (8)$$

where $y_i \in \{0, 1\}$ is the ground-truth that indicates whether slot i corresponds to the referring action $\mathcal{P}^a$. Besides, the demonstration trajectory $\hat{\mathcal{A}}$ is first down-sampled to yield the sparse anchors label $\hat{\mathcal{A}}'$. Using the down-sampling indices, $\hat{\mathcal{A}}$ is then split into N_1 contiguous segments $\{\hat{\mathcal{A}}_i\}_{i=1}^{N_1}$, which serve as labels for LDH supervision. The loss

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}[\|\mathcal{A}' - \hat{\mathcal{A}}'\|_2^2] + \gamma \mathbb{E}[\|\mathcal{A}_i - \hat{\mathcal{A}}_i\|_2^2] \quad (9)$$

jointly supervises GDH and LDH, where the scalar γ balances their relative importance. During training, LDH is updated by uniformly sampling one segment index i per iteration. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{CCE}} + \alpha \mathcal{L}_{\text{MSE}} \quad (10)$$

with the scalar hyper-parameter α balancing the two terms.

4. Experiments

4.1. Referring-Awareness Evaluation

Baselines. To evaluate how well ReV responds to external sparse referring, we select three representative visuomotor policies—ACT [46], DP3 [44] and CDP [15]—as baselines, and compare against them without any architectural change. For each baseline, the referring point $\mathcal{P}$ is injected through its native “condition-and-generate” paradigm: we simply concatenate $\mathcal{P}$ to its original conditioning vector, expecting the model to output trajectories that satisfy the *Manipulation Steering Choreography*. All models are trained on the same expert demonstrations for an identical number of epochs to ensure experimental parity.

Evaluation Metrics. According to the *Manipulation Steering Choreography*, we derive the following three metrics, each averaged over M independent roll-outs.

- **Region Penetration Rate (RePR)** measures the fraction of trajectories in which the robot’s end-effector passes through the referring point $\mathcal{P}$. For trajectory i , let

$$d_i = \min_{0 < t \leq N} \|\mathbf{p}_i(t) - \mathcal{P}\|_2 \quad (11)$$

denote the minimum Euclidean distance between the end-effector position $\mathbf{p}_i(t)$ and $\mathcal{P}$ over the entire trajectory ($0 < t \leq N$). We count a penetration if $d_i \leq \epsilon$ and define

$$\text{RePR} = \frac{1}{M} \sum_{i=1}^M \mathbb{I}[d_i \leq \epsilon] \quad (12)$$

The threshold ϵ specifies the penetration tolerance, and in our experiments it is fixed at 0.05 m.

- **Success Rate (SuR).** In contrast to [15, 30, 44], we assess task success under the steering-fidelity requirement of *Manipulation Steering Choreography*: a roll-out is considered successful only if the robot completes the assigned task and its end-effector passes through the designated referring point $\mathcal{P}$ during execution, i.e.,

$$\text{SuR} = \frac{1}{M} \sum_{i=1}^M \mathbb{I}[d_i \leq \epsilon \wedge S_i = 1] \quad (13)$$

where $S_i \in \{0, 1\}$ is the annotated task-completion label.

- **Smoothness Score (SmS).** For trajectory i , we compute

$$J_i = \frac{1}{N-1} \sum_{t=1}^{N-1} \|\mathbf{p}_{i,t+1} - \mathbf{p}_{i,t}\|_2 \quad (14)$$

To map the unbounded J_i to $[0, 1]$, we use

$$s_i = \exp(-J_i/\lambda) \quad (15)$$

Table 1. **Quantitative results on simulated benchmark**, highlighting the effectiveness of our ReV in referring-aware manipulation.

Method	Pick Meat-via			Lift Barrier-via			Place Food-via			Camera Alignment-via		
	RePR (%) ↑	SuR (%) ↑	SmS ↑	RePR (%) ↑	SuR (%) ↑	SmS ↑	RePR (%) ↑	SuR (%) ↑	SmS ↑	RePR (%) ↑	SuR (%) ↑	SmS ↑
ACT [46]	2%	1%	0.9890	1%	1%	0.9904	0%	0%	-	0%	0%	-
DP3 [44]	80%	1%	0.9899	99%	25%	0.9945	1%	1%	0.9883	0%	0%	-
CDP [15]	14%	14%	0.9924	99%	99%	0.9933	47%	33%	0.9878	0%	0%	-
ReV (Ours)	100%	56%	0.9749	100%	100%	0.9802	100%	50%	0.9812	100%	63%	0.9722

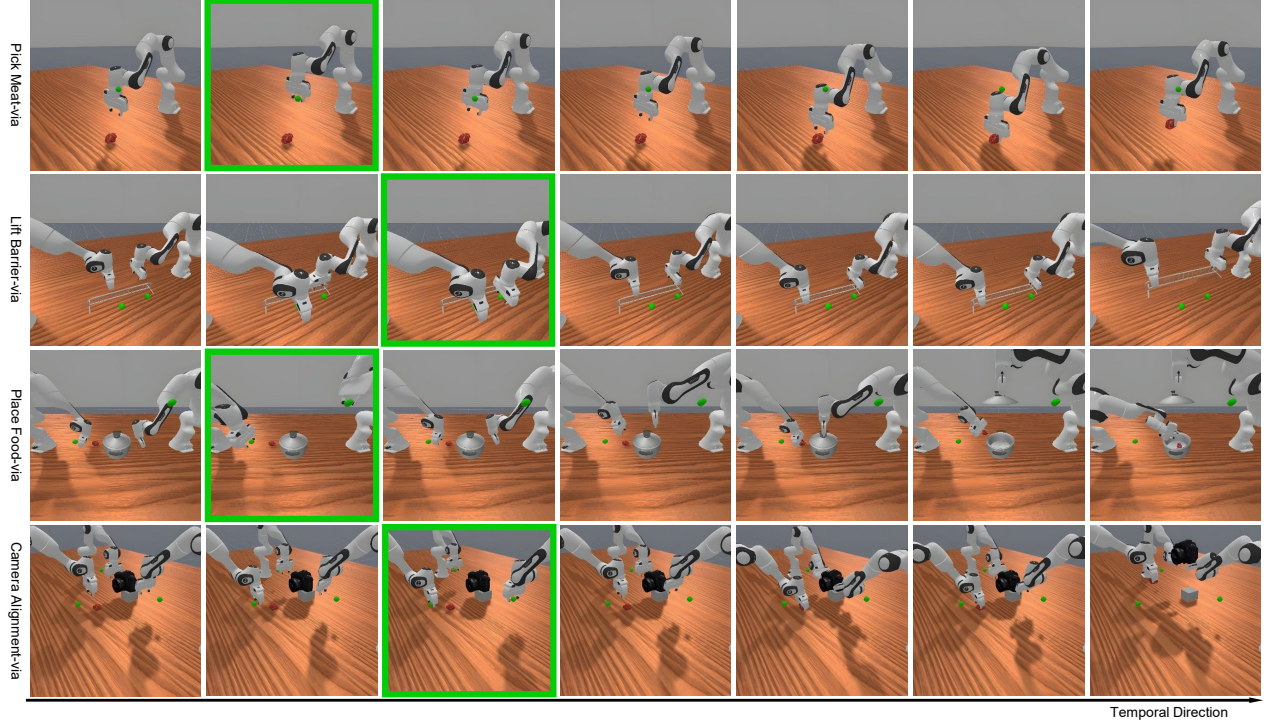


Figure 4. **Visualization of the trajectories generated by ReV** on *Pick-Meat-via*, *Lift-Barrier-via*, *Place-Food-via* and *Camera-Alignment-via*. Here, we use green bounding boxes to mark the frames in which the end-effector passes through the designated via-point (green ball).

with temperature $\lambda > 0$, thus smooth trajectories yield $s_i \approx 1$ and jittery ones $s_i \approx 0$. Subsequently, we define

$$\text{SmS} = \frac{1}{M'} \sum_{i=1}^{M'} s_i \quad (16)$$

where M' denotes the number of roll-outs that satisfy the success criterion used in the definition of SuR.

Benchmarks. We benchmark on four representative tasks from RoboFactory [24]: *Pick Meat*, *Lift Barrier*, *Place Food* and *Camera Alignment*. To meet the *Manipulation Steering Choreography* requirement, we augment each task with a via-point that the robot’s end-effector must traverse en route to successful completion, yielding the modified benchmark *Pick Meat-via*, *Lift Barrier-via*, *Place Food-via* and *Camera Alignment-via*. The via-point generation strat-

egy is detailed in the supplementary material.

Quantitative and Qualitative Results. Tab. 1 reports the quantitative results of all involved methods on our modified benchmark. ReV yields the highest proportion of trajectories that satisfy *Manipulation Steering Choreography*. Thanks to the trajectory-steering strategy, our ReV guarantees that 100% of roll-outs pass through the designated referring point $\mathcal{P}$ (cf. RePR); and its success rate SuR is also influenced by the capability of policy model (see Tab. 2). In contrast, the “condition-and-generate” paradigm adopted by the baselines overwhelmingly ignores $\mathcal{P}$ and proceeds directly to the final goal, exposing its weakness in referring-aware manipulation. Fig. 4 overlays representative trajectories produced by our ReV across the aforementioned tasks, which simultaneously accomplishes the task

Table 2. Quantitative results across multiple simulated benchmarks, highlighting the effectiveness of our policy model.

Method	Adroit [25]		DexArt [3]			MetaWorld [42]				RoboFactory [24]			
	Pen	Door	Laptop	Toilet	Bucket	Reach	Soccer	Sweep Into	Shelf Place	Pick Meat	Lift Barrier	Place Food	Camera Alignment
ACT [46]	47%	66%	35%	8%	6%	21%	28%	23%	43%	16%	20%	3%	7%
DP3 [44]	53%	69%	81%	65%	32%	28%	30%	15%	37%	81%	90%	30%	87%
CDP [15]	68%	74%	84%	68%	32%	22%	23%	24%	35%	84%	93%	51%	90%
ReV (Ours)	73%	79%	87%	71%	61%	36%	33%	27%	47%	94%	99%	57%	94%

while smoothly traversing $\mathcal{P}$, demonstrating markedly superior referring-awareness.

4.2. Effectiveness of Policy Model

Baselines. Following the protocol in Sec. 4.1, we retain ACT, DP3 and CDP as baselines. In this experiment, we withhold $\mathcal{P}$ from all methods—including our ReV—in order to evaluate the intrinsic capability of our policy model with coupled diffusion heads against the myopic window-sliding paradigms employed by the baselines. All involved methods are trained on an identical set of expert demonstrations for the same number of epochs.

Benchmarks. Following [15, 30, 44], we curate a cross-benchmark suite that spans Adroit [25], DexArt [3], MetaWorld [42], and RoboFactory to evaluate the effectiveness of our policy model in various aspects: gripper-based and dexterous manipulation, articulated and rigid objects manipulation, and single- and multi-agent cooperation.

Quantitative Results. Tab. 2 quantitatively shows that our framework consistently outperforms baselines across a diverse set of grasping tasks. These results highlight the decisive importance of the global execution pattern for successful task completion: 1) by modeling long-range dependencies, our policy model ensures consistency along the entire generated trajectory; 2) this strategy endows the model with a macroscopic understanding of task-specific execution motion, thereby avoiding failure cases caused by falling into locally ambiguous robot states.

4.3. Ablation Study

Total Trajectory Length. As noted in Sec. 3.2, our ReV generates *fixed-length* trajectories of N , which must be selected a priori for each task. To quantify sensitivity to this choice, we sweep N across tasks of increasing complexity, and the results are summarized in Fig. 5a. Quantitatively, an overly short N prevents our ReV from capturing the complete execution pattern required for each task, whereas an excessively long N injects redundant and ineffective information during training, degrading performance.

Trajectory-Length Allocation. The total trajectory length N is determined by the sparse-action horizon N_1 produced by GDH and the interpolation horizon N_2 inserted by LDH. We perform an ablation to quantify the sensitivity of ReV

to the balance between these two components while keeping the overall length N fixed for each task. As shown in Fig. 5b, performance peaks around balanced ratios (*i.e.*, $1 : 2 \sim 2 : 1$), whereas larger imbalances consistently degrade results. We attribute this decline to two capacity-mismatch effects: 1) *GDH overload*. A large $N_1 : N_2$ ratio places the burden of modeling complex execution patterns almost entirely on GDH, making it difficult to fit the highly non-linear action manifold. 2) *Supervision dilution*. When the trajectory is over-segmented (*i.e.*, N_2 is too small), the differences between adjacent segments fall below the noise floor. Although LDH employs a learnable, temporal-dependent interpolation strategy, the regression targets become vanishingly simply and the network cannot learn meaningful interpolation kernels for each temporal position.

Temporal-Position Prediction. In the preceding experiments, the temporal-position prediction module is invoked only once at the beginning of inference. Although it could in principle be updated every inference step (Fig. 3), re-computing the temporal position for the same $\mathcal{P}$ repeatedly causes the anchor to drift, producing unstable trajectories. To quantify this effect, we ablate this module by invoking it every i steps ($i \in \{1, 2, 4, N\}$). Fig. 5c shows that the more often we re-predict the position for an *unchanged* $\mathcal{P}$, the larger the performance drop. However, this does not imply that our model is unable to cope with moving or newly-appearing referring points. Whenever the referring point changes—either because objects move or because new ones appear—we simply reset the sparse anchor history $\mathcal{A}'_-$ and restart trajectory generation from the current observation. In this way, ReV immediately adapts to the new configuration without suffering from anchor jitter.

4.4. Real-World Experiments

Platform. We conducted the real-world experiments with a dual-arm setup composed of two ORBBEC PiPER 6-DOF lightweight manipulators, each fitted with a two-finger gripper. An externally mounted, top-down ORBBEC DaBaiDC1 RGB-D sensor delivered a global view of the workspace. Demonstrations were collected through two factory PiPER Teach Pendants that allow simultaneous teleoperation of both arms. A single workstation (NVIDIA GeForce RTX 4090) handles the entire data pipeline:

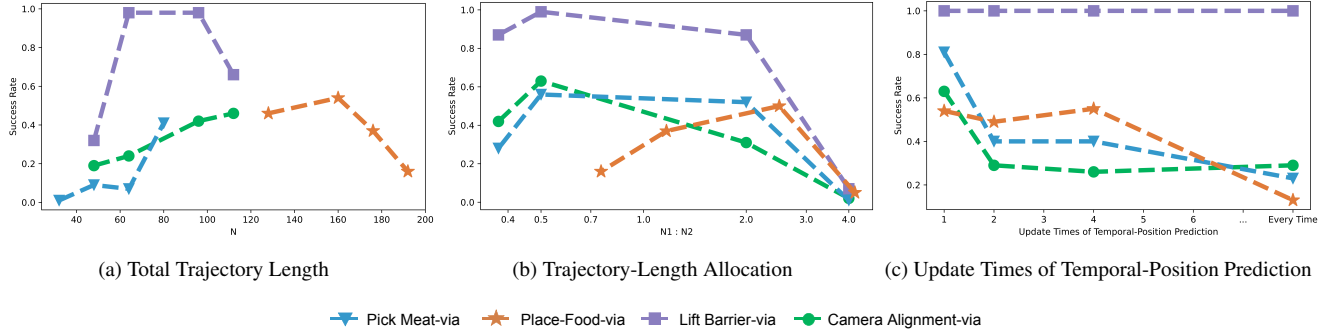


Figure 5. **Visualization of ablation studies on key hyperparameters.** (a) *Total trajectory length N* : performance peaks when N matches task complexity, enabling the model to capture task-specific execution patterns. (b) *Trajectory-length allocation $N_1 : N_2$* : balanced ratios ($1 : 2 \sim 2 : 1$) allow our policy model to learn the entire trajectory most effectively. (c) *Update times of temporal-position prediction*: excessive updates cause anchor drift and degrade performance. Here, success rate is defined as the fraction of trials that simultaneously achieve task completion and pass through the designated referring point, and all ablations were conducted under controlled variables.

Table 3. **Quantitative results on real-world tasks**, highlighting the robustness of our ReV in real-world settings.

Method	Collecting Objects-via				Moving Playing Card Away-via				Grabbing Rod-via		
	PeRP $\uparrow$	GraSuR $\uparrow$	PlaSuR $\uparrow$	SuR $\uparrow$	PeRP $\uparrow$	GraSuR $\uparrow$	PlaSuR $\uparrow$	SuR $\uparrow$	PeRP $\uparrow$	GraSuR $\uparrow$	SuR $\uparrow$
ACT	3 / 30	13 / 30	10 / 13	1 / 30	2 / 30	15 / 30	8 / 15	2 / 30	2 / 30	6 / 30	0 / 30
DP	6 / 30	18 / 30	15 / 18	5 / 30	8 / 30	20 / 30	18 / 20	5 / 30	3 / 30	10 / 30	1 / 30
ReV (Ours)	30 / 30	22 / 30	20 / 22	20 / 30	30 / 30	25 / 30	25 / 25	25 / 30	30 / 30	18 / 30	18 / 30

recording observations, performing policy inference and streaming commands to the arm controllers at 30 Hz.

Demonstrations. The demonstrations utilized in our real-world experiments were generated by teleoperating the dual-arm system with the PiPER Teach Pendants. For each task, we collected a total of 50 demonstrations, each carefully selected to explicitly exhibit the salient motion and object contacts required for reliable success; episodes that deviated from these criteria were discarded and re-recorded.

Tasks. Following Sec. 4.1, we inject a mandatory via-point into the original *Collecting Objects*, *Moving Playing Card Away*, and *Grabbing Rod* tasks to assess the referring-awareness of all involved methods. The resulting tasks are termed *Collecting Objects-via*, *Moving Playing Card Away-via*, and *Grabbing Rod-via*.

Quantitative Results. For each task we constructed a fixed set of 30 diverse real-world trials, and every model was evaluated on the same 30-trial split, ensuring identical test conditions for all comparisons. Tab. 3 lists the overall success rate (SuR) and three constituent metrics: 1) *Region Penetration Rate* (PeRP)—quantifying referring-awareness by checking whether the end-effector passes through the designated referring point; 2) *Grasping Success Rate* (GraSuR)—the percentage of trials in which the robot successfully grasps the target object; 3) *Placing Success Rate* (PlaSuR)—computed only when grasping suc-

ceeds. The overall success rate jointly considers penetration success and final task completion. From this table, we yield that ReV outperforms all baselines, demonstrating the effectiveness of our approach.

5. Conclusion

In this paper, we introduce referring-aware visuomotor policy, a novel close-loop scheme to effectively respond to the external referring information provided by humans or high-level reasoning planners. It robustly handles out-of-distribution perturbations in dynamic environments, only trained by using expert demonstrations under the imitation learning framework without any fine-tuning in the post-processing. To realize this, a carefully designed policy model with coupled diffusion heads is leveraged to generate the detailed action trajectory progressively. Extensive simulated and real-world experiments show that our method outperforms baselines in referring-aware manipulation. In future work, ReV can be coupled with VLMs and world models to tackle more complex and flexible tasks, thanks to their awareness of the reasoning signals.

Limitation. The proposed ReV supports the use of multiple referring points. However, all experiments in this paper only focus on a single referring point due to the naive architecture design of the temporal position prediction module. we will fix it in the future.

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